

Order parameter and mechanism of the RBLapSum through_rank_kappa instability

Research note: one-step score-scale dynamics

2026-09-16/17. Follow-up to the kappa stability investigation.

Contents

1	Purpose and summary of conclusions	1
2	Definitions	3
3	Experiments performed	6
4	Results 1: reparameterization test (H1)	8
5	Results 2: surrogate gain Π (H2)	9
6	Results 3: signed drift and second-order heating (H3)	10
7	Results 4: counterfactual interventions and collapse dynamics (H4)	14
8	Results 5: prospective comparison of candidate indicators	16
9	Results 5b: surrogate-gain fate map	17
10	Results 6: drift, heating, and the escape picture	17
11	Proposed causal sequence for a burst	18
12	Recommended controller variables	19
13	Remaining uncertainties and next experiments	20

1 Purpose and summary of conclusions

The previous investigation (`docs/rblapsum-kappa-stability-neutral.tex`) introduced the geometric quantity

$$\chi_{\text{geo}} = \frac{b}{2T\delta}$$

as an empirical indicator of instability.

The purpose of the present investigation is to test whether χ_{geo} is a fundamental control parameter or only a useful proxy, and to identify the mechanism that produces the observed score-inflation bursts.

Four hypotheses are considered:

- **H1:** χ_{geo} is an empirical proxy rather than a fundamental order parameter.
- **H2:** the dimensionless surrogate gain

$$\Pi = \|L_{\kappa}D_z\|_F^2$$

is a more appropriate measure of the strength of the support-gradient mechanism.

- **H3:** large surrogate gain destabilizes the model by coupling to the score-scale degree of freedom.
- **H4:** the collapse of the effective kernel population, $n_{\text{eff}} \rightarrow 1$, is primarily a late-stage feature of the instability rather than its initial cause.

The main results are as follows.

- **H1 is supported.** A score-unit reparameterization leaves the forward computation and training dynamics unchanged while changing χ_{geo} by an arbitrary multiplicative factor. In addition, within a fixed (K, T) row, χ_{geo} has essentially no prospective predictive power for future bursts (AUC approximately 0.46). Therefore a universal numerical threshold in χ_{geo} cannot be fundamental.
- **H2 is supported as a statement about surrogate gain.** The quantity

$$\Pi = \|L_{\kappa}D_z\|_F^2$$

is dimensionless and invariant under the score-unit reparameterization. Across the measured states it predicts the realized gradient gain with rank correlation 0.97, compared with 0.52 for χ_{geo} . The approximation

$$\Pi \approx \frac{b^2}{4T\delta}$$

is reasonably accurate in healthy, locally dense boundary regions, but can fail badly when the score geometry becomes degenerate.

- **H3 is supported, but the dominant mechanism is not a positive first-order radial force.** At the dangerous states, the signed task and surrogate contributions to the score scale are generally restoring. The destabilizing contribution is instead a second-order increase in score scale caused by large parameter updates. We refer to this contribution as *heating*. It is even under reversal of the gradient direction and scales approximately quadratically with the learning-rate multiplier. At the most dangerous states it is largely deterministic rather than a consequence of minibatch variance. Instability begins when this heating becomes comparable to or larger than the restoring signed drift.
- **H4 is supported with an important qualification.** In the strict one-feature limit, $n_{\text{eff}} = 1$, the kappa-corrected surrogate vanishes: the measured Π is exactly zero. However, neighboring blocks with $n_{\text{eff}} \approx 2$ and very large activations can retain enormous surrogate gain and continue to increase the score scale. Post-collapse growth stops when the support gradient is disabled, while broadening the kernel after collapse can make the instability substantially worse.

- **The resulting controller picture has two relevant variables.** The effective kernel population n_{eff} provides the strongest measured within-row early warning of a burst, while Π measures the strength of the surrogate coupling. After a block has already entered the collapsed regime, the appropriate emergency intervention is to suppress the support gradient rather than to increase T .

2 Definitions

All gate quantities below are computed per layer from the candidate set, consisting of the top $K + J$ features by score.

Scores, boundary, and kernel

The pre-gate feature value is z_i , and the score used for ranking is

$$s_i = |z_i|.$$

The rank boundary is

$$b = \max(b_0, s_{(K+1)}),$$

where $s_{(K+1)}$ is the largest inactive score.

The exponential kernel is

$$\kappa_i = \frac{e^{-|s_i - b|/T}}{2T}, \quad Z = \sum_i \kappa_i.$$

The effective kernel population is

$$n_{\text{eff}} = \frac{Z^2}{\sum_i \kappa_i^2}.$$

A local rank-spacing estimate is

$$\delta = \frac{s_{(K-3)} - s_{(K+5)}}{8}.$$

Small δ corresponds to a locally dense score distribution near the support boundary.

Gradient chain

It is useful to make the backward path explicit. Let p_i denote the surrogate gate variable and let the bottleneck output be

$$y_i = p_i z_i.$$

The gradient arriving from the rest of the network is

$$u_i = \frac{\partial L}{\partial y_i}.$$

At fixed z_i ,

$$\frac{\partial y_i}{\partial p_i} = z_i,$$

so

$$\frac{\partial L}{\partial p} = D_z u, \quad D_z = \text{diag}(z).$$

For the kappa-corrected surrogate, define

$$D_\kappa = \text{diag}(\kappa)$$

and

$$L_\kappa = D_\kappa - \frac{\kappa \kappa^\top}{Z}.$$

The surrogate score-to-gate Jacobian is represented by L_κ . The support-gradient contribution with respect to the scores is therefore

$$g_s = L_\kappa D_z u.$$

Equivalently, with

$$w_i = -u_i z_i,$$

one may write

$$g_s = -L_\kappa w,$$

up to the sign convention used for the reported update direction.

Because

$$L_\kappa \mathbf{1} = 0,$$

the kappa correction removes the common score-translation mode from the surrogate gradient.

Exact surrogate gain

The dimensionless gain of the surrogate Jacobian is defined as

$$\boxed{\Pi = \|L_\kappa D_z\|_F^2}.$$

Using the diagonal-minus-rank-one form of L_κ , this can be computed without constructing a dense matrix:

$$\Pi = \sum_j z_j^2 \kappa_j^2 \left[\left(1 - \frac{\kappa_j}{Z}\right)^2 + \frac{\sum_i \kappa_i^2 - \kappa_j^2}{Z^2} \right].$$

The quantity $\kappa_j z_j$ is dimensionless, so Π is also dimensionless.

By contrast,

$$\chi_{\text{geo}} = \frac{b}{2T\delta}$$

has units of inverse score.

For a locally dense and approximately uniform score distribution near the boundary,

$$\Pi \approx \frac{b^2}{4T\delta}.$$

This approximation is tested below rather than assumed to hold generally.

The realized empirical gain is measured as

$$G_{\text{emp}} = \frac{\|g_s\|^2}{\|u\|^2}$$

over the candidate features.

Score-scale statistics

On a fixed evaluation batch, three scale statistics are monitored:

$$R_{\text{all}} = \text{RMS}_i(s_i),$$

the mean top- K score R_K , and the boundary b .

The logarithmic scale coordinate is

$$x = \log R.$$

Using $\log R$ makes multiplicative changes in score scale additive.

One-step drift and heating decomposition

Counterfactual optimizer steps are evaluated from the same saved model and optimizer state.

The branches are:

- *null*: zero current gradient;
- *task*: support-surrogate gradient disabled;
- *full*: ordinary task gradient plus support-surrogate gradient;
- the corresponding antithetic branches in which the current gradient is reversed.

The saved Adam-type optimizer state is retained in every branch, so previously accumulated momentum and second moments are replayed identically.

For a gradient G , define the signed, or odd, component of the one-step scale change as

$$\mu(G) = \frac{1}{2} \mathbb{E} [\Delta x(+G) - \Delta x(-G)].$$

This isolates the part that changes sign when the current gradient direction is reversed.

The support contribution to the signed drift is obtained from

$$\mu_{\text{supp}} = \mu_{\text{full}} - \mu_{\text{task}}.$$

The even component relative to the null branch measures the sign-independent change in scale. At second order, parameter displacements contribute positively to squared activation norms even when their first-order radial contribution changes sign. This even contribution is referred to below as *heating*.

The raw gradients themselves satisfy the exact decomposition

$$G_{\text{full}} = G_{\text{task}} + G_{\text{supp}},$$

which was verified using identical random-number states and in a model with no stochastic layers. Optimizer updates are not additive in the same way, because the Adam-type update is nonlinear in the current gradient and optimizer state. For this reason the optimizer branches are measured separately.

3 Experiments performed

The investigation used the following experiments.

1. **Checkpoint library.** Thirteen model states were selected to cover several regimes:
 - calm $T = 2$ states;
 - stable $K = 32, T = 1$ states;
 - a marginal surviving $K = 32, J = 224, T = 0.5$ run around a recoverable burst, using checkpoints at steps 2000, 4000, and 6000;
 - marginal checkpoints from runs that later collapsed, including $K = 32, J = 480, T = 0.5$ and $K = 64, J = 448, T = 1$;
 - a post-collapse $K = 32, J = 480, T = 0.1$ state.

All checkpoints include the full optimizer state.

The state labels describe the eventual fate of the corresponding run. Except for the explicitly post-collapse checkpoint, the model is still in a healthy state at the time of measurement. For example, the $K = 64, T = 1$ “pre-collapse” checkpoint is taken at step 2000, while the original run destabilizes at approximately step 3940.

2. **One-step drift and heating.** For each state, 16 batches were evaluated. Five counterfactual optimizer branches were measured per batch, and the per-layer change in $\log R$ was evaluated on a fixed evaluation batch.
3. **Surrogate gain and radial structure.** For each state and layer, the analysis measured

$$\Pi, \quad \frac{b^2}{4T\delta}, \quad \chi_{\text{geo}}, \quad n_{\text{eff}}, \quad G_{\text{emp}},$$

as well as the support radial projection

$$r_{\text{supp}} = -\frac{s^\top g_s^{(z)}}{s^\top s},$$

the quantity

$$(s - b\mathbf{1})^\top L_\kappa w,$$

and

$$\text{Cov}_q(s - b, w).$$

A permutation control was also performed using 48 random shuffles of w among candidates for each token.

4. **Scale-response curves.** A function-preserving gate-level intervention rescales

$$z \rightarrow \alpha z$$

while dividing the gate output by α . This keeps the forward function and ordinary task-gradient path unchanged, while the support surrogate sees scores multiplied by α at fixed temperature T .

The intervention was unit-tested and verified numerically. Drift was measured for

$$\alpha \in \{0.5, 0.75, 1, 1.25, 1.5, 2, 3\}$$

at four reference states.

5. **Score-unit reparameterization.** The transformation

$$z \rightarrow cz, \quad s \rightarrow cs, \quad b \rightarrow cb, \quad T \rightarrow cT,$$

together with the corresponding output compensation, should leave the gate and parameter gradients invariant.

Under this transformation,

$$\chi_{\text{geo}} \rightarrow \frac{\chi_{\text{geo}}}{c}.$$

This provides a direct test of whether a numerical threshold in χ_{geo} can be fundamental.

6. **Learning-rate and batch-size tests.** Learning-rate multipliers

$$\{0.5, 1, 2\}$$

and a half-batch condition were evaluated at three representative states in order to distinguish first-order signed drift from second-order or stochastic effects.

7. **Counterfactual continuations.** Longer continuations were run from a marginal $K = 96, T = 1$ state and from the post-collapse $T = 0.1$ state.

The interventions included:

- normal continuation with three random seeds;
- disabling the support gradient from the start;
- disabling the support gradient after a boundary-inflation trigger;
- increasing T by a factor of 3 in the marginal state;
- increasing T by a factor of 30 in the post-collapse state;

- applying the temperature intervention either immediately or only after the trigger.
8. **Prospective comparison of candidate order parameters.** A dataset of 148 checkpoint states from 18 earlier fixed- T runs was used to test whether a quantity measured at step r predicts a boundary burst during

$$(r, r + H].$$

Predictions were evaluated both after pooling all configurations and within fixed (K, T) rows.

4 Results 1: reparameterization test (H1)

Consider the gate-level transformation

$$(s, z, b, T) \rightarrow (cs, cz, cb, cT).$$

With the corresponding output compensation, this transformation leaves the forward computation and parameter gradients invariant.

This invariance is exact in the float32 single-gate unit tests. It is also reproduced at full checkpoints. For

$$c = 0.5 \quad \text{and} \quad c = 2,$$

which are exactly representable binary rescalings, the training loss is unchanged to all reported digits. The largest relative parameter-gradient difference is approximately

$$2 \times 10^{-3} - 6 \times 10^{-3},$$

consistent with differences in bf16 accumulation order.

For $c = 3$, which is not an exact power-of-two scaling, some nearly tied scores change ordering in bf16. This changes candidate membership and produces a loss shift of order

$$10^{-3}.$$

This effect is attributed to finite-precision ranking rather than to a violation of the underlying reparameterization symmetry.

Under exactly the same transformation,

$$\chi_{\text{geo}} \rightarrow \frac{\chi_{\text{geo}}}{c}.$$

For example, in layer 4 the measured values are

$$54.1 \quad (c = 1),$$

$$27.0 \quad (c = 2),$$

and

$$18.1 \quad (c = 3).$$

In contrast, Π is invariant.

Therefore functionally equivalent gate parameterizations can have different values of χ_{geo} . This rules out a universal numerical instability threshold based directly on χ_{geo} .

The prospective analysis in Section 8 gives a second, independent limitation: within a fixed (K, T) row, χ_{geo} has approximately chance-level predictive performance.

H1 conclusion. The measurements support interpreting χ_{geo} as an empirical configuration-level proxy rather than a fundamental order parameter. It distinguishes some (K, T) regimes because it co-varies with the local kernel geometry, but it is neither reparameterization-invariant nor a useful within-row temporal warning signal.

5 Results 2: surrogate gain Π (H2)

Across 96 layer states (12 checkpoints $\times$ 8 blocks, excluding the post-collapse state), the exact quantity

$$\Pi = \|L_{\kappa} D_z\|_F^2$$

predicts the realized gradient gain

$$G_{\text{emp}} = \frac{\|g_s\|^2}{\|u\|^2}$$

with rank correlation

$$0.97.$$

For comparison, the rank correlation with χ_{geo} is

$$0.52.$$

This strongly supports the interpretation of Π as the relevant surrogate-Jacobian gain.

The dense-boundary approximation

$$\Pi \approx \frac{b^2}{4T\delta}$$

is typically accurate within a factor of approximately two in mid-depth blocks of healthy states.

Its accuracy deteriorates as the local score distribution becomes nonuniform or degenerate. The discrepancy reaches approximately a factor of 10 during late inflation and approximately 10^4 after collapse.

The approximation can also overestimate the gain by orders of magnitude in individual blocks where several rank scores become nearly tied. In that case,

$$\delta \rightarrow 0$$

even though the exact kernel geometry and exact Π remain finite.

This is an important limitation of any order parameter based directly on the single local-spacing statistic δ .

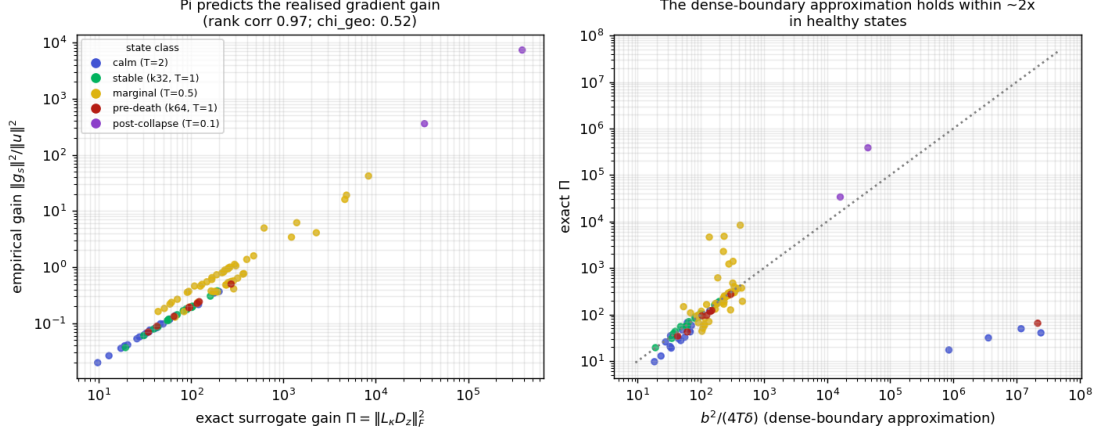


Figure 1: Left: exact surrogate gain Π versus the realized gradient gain $\|g_s\|^2/\|u\|^2$. Right: exact Π versus the dense-boundary approximation $b^2/(4T\delta)$. Each point corresponds to one checkpoint and one block. The checkpoint classes are calm ($T = 2$), stable ($K = 32, T = 1$), marginal ($T = 0.5$), pre-collapse ($K = 64, T = 1$), and post-collapse ($T = 0.1$). The class denotes the eventual fate of the run, not necessarily its condition at the measured checkpoint. In particular, the pre-collapse checkpoint is still healthy at step 2000 and destabilizes approximately 1900 steps later. Only the post-collapse points are measured after a collapse. No averaging over blocks is performed. Most checkpoints are taken at step 2000, with additional $T = 0.5$ states at steps 4000 and 6000 and one calm and one stable checkpoint at step 10000. Within each point, Π is averaged over tokens on the fixed evaluation batch, while G_{emp} is averaged over tokens on one training microbatch.

The earlier comparison between the K - and T -axes is approximately reproduced by the dense-boundary formula. At step 2000,

$$\frac{\Pi_{\text{approx}}(K = 64, T = 1)}{\Pi_{\text{approx}}(K = 32, T = 1)} = 1.66,$$

within the previously predicted range 1.6–2.25.

However, for the exact gain the corresponding ratio is only

$$1.09.$$

The difference arises because the approximation replaces the actual kernel and z -profile by a locally uniform-density model, whereas the exact Π uses the full candidate geometry.

H2 conclusion. The data support Π as the appropriate measure of the instantaneous gain of the support-surrogate Jacobian. It is dimensionless, reparameterization-invariant, and closely predicts the realized surrogate gradient gain.

This does *not* by itself imply that a large Π is unstable. Whether the injected gradient produces score-scale growth is a separate dynamical question.

6 Results 3: signed drift and second-order heating (H3)

The decomposition gives three main results.

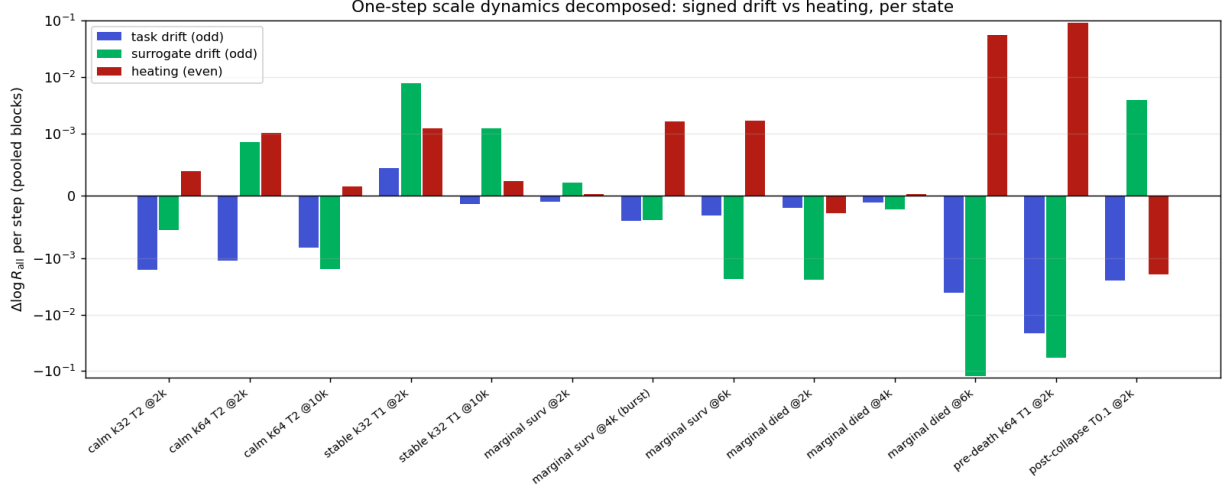


Figure 2: One-step change in $\log R_{\text{all}}$, decomposed into signed task drift, signed support-surrogate drift, and the even heating contribution. The signed terms are obtained from the antithetic gradient branches. Results are pooled over blocks at thirteen checkpoint states.

Signed surrogate drift is not the primary runaway term

At the $K = 64, T = 1$ checkpoint measured approximately 1940 steps before collapse, the signed surrogate contribution is strongly restoring:

$$\mu_{\text{supp}} \approx -5.7 \times 10^{-2}$$

per step when pooled over blocks, and approximately

$$-0.10$$

at the most extreme block.

The ordinary task contribution is also restoring:

$$\mu_{\text{task}} \approx -2.2 \times 10^{-2}.$$

A similar pattern occurs during late inflation in the unstable $T = 0.5$ run.

Thus the instability is not explained by a first-order support force that directly pushes all scores outward. At the measured dangerous states, the signed contribution tends to reduce rather than increase the scale.

The dominant destabilizing contribution is second-order heating

The even component of the update increases the activation scale independently of the sign of the current gradient.

At the pre-collapse $K = 64, T = 1$ state, this contribution reaches

$$+9.2 \times 10^{-2}$$

per step after pooling over blocks, and approximately

$$+0.18$$

at the most strongly affected block.

This exceeds the restoring signed drift.

At calm and stable checkpoints, the same heating term is approximately 30–300 times smaller.

The learning-rate sweep supports the interpretation as a second-order effect: the signed odd terms scale approximately linearly with the learning-rate multiplier, while the even contribution scales approximately quadratically.

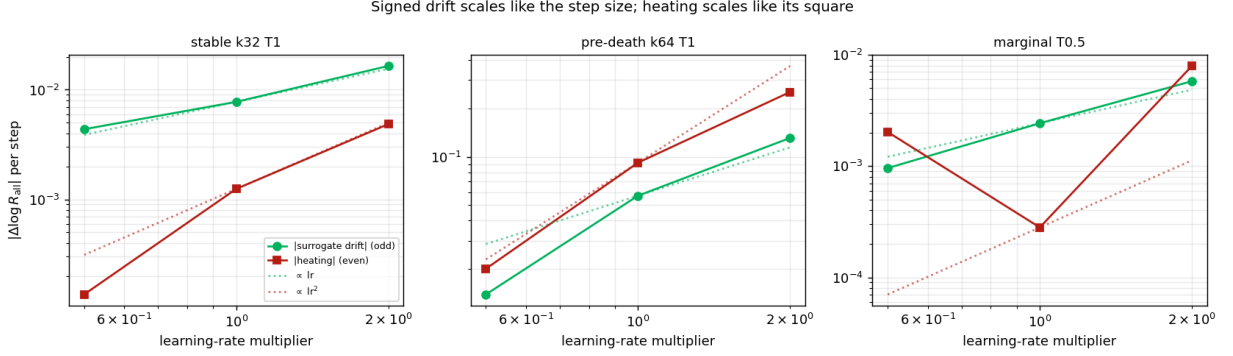


Figure 3: Signed drift and heating as functions of the learning-rate multiplier at three checkpoint states. Reference linear and quadratic scalings are shown for comparison.

Dangerous-state heating is mostly deterministic

At the pre-collapse state, halving the batch size changes the measured heating only from approximately

$$9.2 \times 10^{-2}$$

to

$$9.3 \times 10^{-2}.$$

The variation across batches is also small.

Therefore the large heating at this checkpoint is not primarily caused by minibatch-gradient noise. It is better interpreted as deterministic second-order scale inflation caused by a large structured optimizer step.

At the stable $K = 32, T = 1$ state, the much smaller heating contribution approximately doubles when the batch size is halved. In that regime, stochastic gradient variance appears to make a substantial contribution.

Scale-response curves

The scale-response experiment isolates the signed support component.

At the calm $K = 64, T = 2$ state, the support-surrogate drift crosses zero near

$$\alpha \approx 1.4.$$

Below this point it tends to increase scale, while above it the sign becomes restoring. Within this gate-level intervention, the surrogate therefore behaves approximately like a restoring force around a preferred scale.

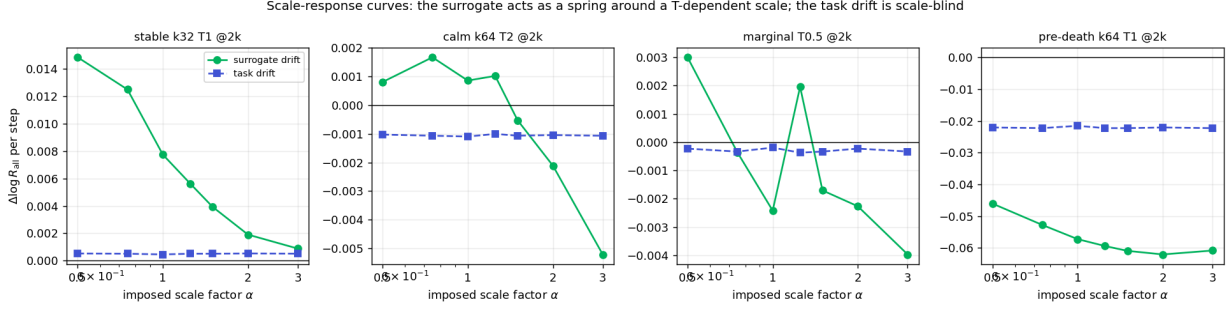


Figure 4: Signed surrogate and task drift as functions of an imposed function-preserving scale factor α at four checkpoint states.

At the stable $K = 32, T = 1$ state, the signed surrogate contribution remains positive throughout the measured range, but decreases in magnitude as α increases.

The task contribution is independent of α by construction of the intervention, and the measurements reproduce this property. This provides an internal validation that the experiment isolates the surrogate scale dependence as intended.

Rank structure of the signed force

The permutation test asks whether the relationship between the task preference w_i and feature rank matters, rather than only the overall magnitude of w .

In most measured states, the true signed radial component differs significantly from the rank-shuffled null distribution, typically by

$$|z\text{-score}| \gtrsim 3\text{--}20.$$

At the pre-collapse state, the alignment is outward relative to the shuffled null,

$$+19.8\sigma,$$

whereas late calm states show inward alignment.

The absolute magnitude of this signed component is nevertheless small compared with the heating term at the dangerous checkpoints.

The data therefore suggest that the rank structure of w determines the signed scale response, which is often restoring, while the main destabilization at high pressure is associated with the even second-order term.

H3 conclusion. Large surrogate gain does not destabilize the model primarily through a positive first-order radial force.

The measurements instead support the following local picture:

$$\text{large support-gradient step} \longrightarrow \text{second-order score-scale heating},$$

while the signed task and support components provide an opposing restoring contribution.

A block becomes locally unstable when the heating contribution becomes comparable to or larger than this restoration.

7 Results 4: counterfactual interventions and collapse dynamics (H4)

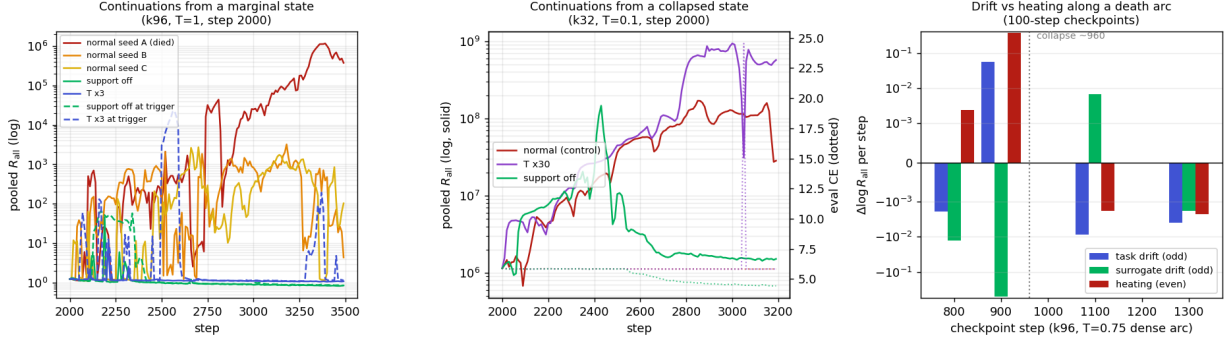


Figure 5: Left: continuations from the marginal $K = 96, T = 1$ state at step 2000, comparing three normal seeds with support-off and $T \times 3$ interventions, applied either immediately or after a boundary-inflation trigger. Middle: continuations from the collapsed $T = 0.1$ state; solid lines show scale and dotted lines show evaluation CE. Right: one-step drift/heating decomposition along a densely sampled $K = 96, T = 0.75$ collapse trajectory, with checkpoints every 100 steps and collapse near step 960.

Interventions from a marginal state

Starting from the $K = 96, T = 1$ checkpoint, three normal 1500-step continuations were run.

All three show repeated score-scale excursions of approximately one to three orders of magnitude. Most excursions recover, but one of the three replicas reaches terminal divergence.

When the support gradient is disabled, the trajectory becomes calm and the score scale decreases. This is consistent with the one-step measurements showing that the ordinary task gradient is restoring at this state.

Increasing the temperature by a factor of three also suppresses the bursts while keeping the scale approximately constant. This branch obtains evaluation CE 1.505, compared with 1.552–1.634 for the normal branches.

Both interventions remain effective when activated only at the first boundary-inflation trigger, defined by the boundary exceeding three times its rolling median.

These experiments show that, for this marginal checkpoint, the support surrogate is necessary for the observed burst dynamics.

Interventions from an already collapsed state

The post-collapse $T = 0.1$ checkpoint begins at a score scale of approximately

$$10^6.$$

Under normal continuation, the scale increases by another factor of approximately 25 over 1200 steps, while CE remains near 5.8.

Increasing the temperature by a factor of 30 does not rescue this state. Instead, the scale grows by approximately a factor of 500.

The reason is visible in the gate geometry. Broadening the kernel repopulates candidate regions in a state where some blocks already have extremely large surrogate gain,

$$\Pi \sim 10^{12}.$$

At the same time, one block is exactly in the single-feature limit,

$$n_{\text{eff}} = 1, \quad \Pi = 0.$$

Thus the strict one-feature limit itself has zero corrected surrogate gain, as predicted by

$$L_{\kappa} \rightarrow 0.$$

Other near-degenerate blocks, however, still contain more than one significantly weighted feature and have very large z . Their surrogate gain can therefore remain extremely large.

Disabling the support gradient stops the continued inflation. The model then begins to recover, with evaluation CE decreasing from

$$5.93$$

to

$$4.44$$

over 1200 steps.

Therefore post-collapse inflation in this experiment is not maintained by the task gradient alone. It requires the remaining support-surrogate contribution.

Dense sampling of a collapse

A separate $K = 96, T = 0.75$ trajectory was sampled every 100 steps around its collapse near step 960.

Approximately 200 steps before collapse, the one-step decomposition is still moderate:

$$\text{restoring drift} \approx -1.3 \times 10^{-2},$$

$$\text{heating} \approx +2.4 \times 10^{-3}.$$

The heating is already approximately an order of magnitude larger than in calm states, but restoration still dominates.

Around 60–100 steps before collapse, both contributions increase by roughly two orders of magnitude:

$$\text{surrogate signed drift} \approx -0.50,$$

$$\text{heating} \approx +0.37$$

per step after pooling over blocks.

The model is therefore not experiencing a simple unopposed outward force. Instead, very large restoring and destabilizing contributions coexist, and their balance differs across layers.

After collapse, both measured terms become small again because much of the kernel has effectively disappeared.

H4 conclusion. The measurements support treating $n_{\text{eff}} \rightarrow 1$ as a late-stage consequence of the instability rather than its initial trigger.

The exact $n_{\text{eff}} = 1$ limit eliminates the corrected support gradient. However, the approach to this limit is not harmless: blocks with $n_{\text{eff}} \approx 2$ and very large activations can retain extremely large surrogate gain and continue to drive scale growth.

Thus two observations are simultaneously true:

1. the distributed character of the correction is lost as the kernel population collapses;
2. the residual surrogate in neighboring near-degenerate blocks can remain strongly destabilizing until it is disabled or the kernel becomes fully inactive.

8 Results 5: prospective comparison of candidate indicators

The candidate quantities were evaluated prospectively on 148 checkpoint states from 18 earlier fixed- T runs. Post-collapse checkpoints were excluded.

For each checkpoint at step r , the target variable is whether a boundary-inflation burst occurs during

$$(r, r + H].$$

The results are:

H	pooled AUC			within-row AUC		
	χ_{geo}	Π_{approx}	$1/n_{\text{eff}}$	χ_{geo}	Π_{approx}	$1/n_{\text{eff}}$
1000	0.81	0.88	0.77	0.46	0.63	0.75
2000	0.78	0.86	0.71	0.48	0.68	0.75
4000	0.80	0.85	0.68	0.44	0.69	0.76

When all configurations are pooled, Π_{approx} gives the best predictive performance.

This comparison, however, includes large differences between (K, T) configurations. A more informative test of temporal warning within a given operating regime is obtained by evaluating AUC separately within fixed (K, T) rows.

Under this comparison, χ_{geo} falls to approximately chance level:

$$\text{AUC} \approx 0.44\text{--}0.48.$$

The strongest within-row predictor is low effective kernel population:

$$1/n_{\text{eff}},$$

with AUC approximately

$$0.75\text{--}0.76.$$

The approximate surrogate gain remains informative but is weaker, with AUC approximately

$$0.63\text{--}0.69.$$

Only three marginal (K, T) rows contribute both positive and negative outcomes to this within-row comparison. The numerical AUC values should therefore be viewed as preliminary rather than definitive.

Nevertheless, the qualitative distinction is clear: χ_{geo} mainly separates different operating regimes, while decreasing kernel population contains additional temporal information about an approaching burst.

9 Results 5b: surrogate-gain fate map

The earlier dose-response analysis was repeated using the new surrogate-gain quantities and resolving the results by block and training checkpoint.

Only pre-collapse checkpoints are included. The trajectory panel follows the block with the largest gain in each run over training. The dose-response panel compares the pre-onset burst rate with the earliest healthy value of the worst-block gain.

The current dataset includes eighteen fixed- T campaign runs with checkpointed score ladders for

$$K = 32, 64.$$

A separate boundary-bracketing campaign over

$$K = 32\text{--}128$$

and sixteen configurations was not yet complete at the time of this report.

10 Results 6: drift, heating, and the escape picture

The learning-rate and batch-size experiments distinguish two contributions to the scale dynamics. The signed drift scales approximately linearly with the learning-rate multiplier.

The even heating contribution scales approximately quadratically with the learning-rate multiplier. At the stable $K = 32, T = 1$ state, halving the batch size approximately doubles the heating contribution. This indicates a substantial stochastic-gradient component in that regime.

At the pre-collapse $K = 64, T = 1$ state, halving the batch leaves the heating essentially unchanged. The dominant heating there therefore appears to be deterministic second-order overshoot from a large structured update.

The earlier observation of stochastic-looking score excursions is reproduced directly by the marginal continuation experiment: among three replicas from the same marginal state, one reaches terminal collapse while two undergo bursts and recover.

These observations are consistent with a metastable escape picture, but the available measurements do not yet support a quantitative reduction to a single Kramers-like escape parameter.

For example, at the calm $K = 64, T = 2$ state, the restoring scale-response curve crosses zero near

$$\alpha \approx 1.4,$$

with an estimated local slope

$$\lambda \approx 7 \times 10^{-3}$$

per unit $\log R$ per training step.

Comparable estimates of the restoring slope, effective excitation, and distance to the collapse boundary are not yet available across enough marginal configurations.

The qualitative picture supported by the current data is therefore:

$$\text{restoring scale dynamics} + \text{second-order excitation} + \text{kernel-population collapse}.$$

The task gradient generally supplies a restoring contribution. The support surrogate can also provide a signed restoring contribution around its scale-dependent operating point.

At the same time, large support-induced parameter updates generate second-order heating. If an excursion becomes sufficiently large, the effective kernel population decreases and the distributed correction enters a near-degenerate regime.

This provides a mechanism for recoverable bursts as well as rare terminal escapes without requiring a persistent positive first-order radial force.

11 Proposed causal sequence for a burst

The following sequence summarizes the present interpretation. Statements directly established by the measurements are marked **(M)**; steps that remain interpretive are marked **(I)**.

1. In marginal configurations, the support surrogate has substantially larger gain than in calm configurations: the measured Π is approximately 3–10 times larger. The resulting optimizer step produces second-order heating that is comparable to, or exceeds, the restoring signed drift in the most affected blocks. **(M)**
2. This heating can produce temporary increases in score scale. Both the task and signed support contributions generally oppose the scale increase at the measured dangerous states. Most observed excursions therefore recover. **(M)**
3. During a sufficiently large excursion, the effective kernel population decreases. This is observed directly in the $T = 0.1$ collapse trajectory, and low n_{eff} is the strongest prospective within-row predictor of future bursts. **(M)**

As the population becomes small, the correction is concentrated among fewer features, potentially increasing the gain carried by each remaining nondegenerate direction. **(I)**

4. In the exact single-feature limit,

$$n_{\text{eff}} = 1,$$

the corrected support Jacobian vanishes and

$$\Pi = 0.$$

This limit is observed in one post-collapse block. At the same time, neighboring blocks with $n_{\text{eff}} \approx 2$ and very large z can have

$$\Pi \sim 10^{12}.$$

Disabling the support gradient stops post-collapse scale growth, showing that these remaining surrogate contributions are causally important. **(M)**

5. Once several blocks have entered the high-scale regime, scale amplification propagates through depth in the residual-replacement architecture, as observed in the earlier investigation. The trajectory eventually reaches a high-scale state in which most of the kernel is effectively inactive. (M)

12 Recommended controller variables

The experiments suggest separating two functions that were previously combined in the χ_{geo} -based controller.

Fast warning variable: effective kernel population

The strongest prospective within-row warning signal is the effective population

$$n_{\text{eff}}.$$

Low n_{eff} reaches within-row AUC approximately 0.75, and population collapse is directly associated with entry into the near-degenerate regime.

The existing temperature controller already contains a related population floor based on the number of candidates occupying the kernel window. The present results support keeping such a fast population-based guard.

However, the intervention should depend on the stage of the instability.

Before collapse, broadening the kernel can restore a healthy distributed correction.

After a block has already entered the extremely high-scale, near-degenerate regime, increasing T can repopulate the kernel while z is enormous, producing very large surrogate gain. The post-collapse intervention shows that this can accelerate rather than suppress the runaway.

At that stage, the measured safe emergency action is to disable or strongly reduce the support-gradient contribution until the scale has recovered.

Pressure variable: exact surrogate gain

The natural measure of support-surrogate strength is

$$\Pi = \|L_{\kappa} D_z\|_F^2.$$

It is dimensionless, invariant to score-unit reparameterization, and closely tracks the realized support-gradient gain.

When the exact candidate tensors are unavailable, the forward-only approximation

$$\Pi_{\text{approx}} = \frac{b^2}{4T\delta}$$

provides a useful coarse proxy, although it becomes unreliable near degenerate score geometries.

A controller based directly on Π would therefore be a more principled replacement for the current χ_{geo} -based pressure trim.

The existing χ_{geo} controller works in the tested regime because χ_{geo} co-varies with surrogate gain across configurations, but χ_{geo} itself has no invariant interpretation and should be treated as a configuration-level heuristic.

Direct dynamical indicator

The quantity most directly related to imminent scale instability is not Π or n_{eff} alone, but the competition between heating and restoration in the most affected block.

The antithetic one-step probe measures both terms directly.

A useful diagnostic ratio is schematically

$$\mathcal{R}_{\text{heat}} = \frac{\text{second-order heating}}{|\text{restoring signed drift}|}.$$

The measured values are approximately

$$\mathcal{R}_{\text{heat}} \approx 0.02$$

at calm states,

$$\mathcal{R}_{\text{heat}} \approx 0.7\text{--}1.8$$

at states whose runs collapse within approximately 2000 steps, and

$$\mathcal{R}_{\text{heat}} \sim 2$$

roughly 100 steps before collapse.

Thus the direct dynamical criterion suggested by the measurements is

$\mathcal{R}_{\text{heat}} \gtrsim 1$

heating becomes comparable to or larger than restoration.

The practical forward-only indicators are the pair

$$(\Pi, n_{\text{eff}})$$

evaluated per block.

Here Π measures the squared gain of the surrogate Jacobian. At fixed upstream-gradient scale and optimizer response, a larger Π therefore tends to increase the squared parameter-step magnitude, which is the quantity relevant to second-order heating.

This point is worth stating carefully: heating is quadratic in the *parameter-step amplitude*. Because Π is already a squared gain, one should not generally expect heating to scale as Π^2 . Under otherwise fixed conditions, a roughly linear relation between heating and Π is the more natural leading expectation.

The second quantity, n_{eff} , measures how close the kernel is to the near-degenerate regime where the distributed correction is lost.

Together these variables provide a mechanistically clearer description than χ_{geo} :

Π measures support-surrogate gain, n_{eff} measures proximity to kernel degeneracy.

13 Remaining uncertainties and next experiments

Several important questions remain open.

- The destabilizing heating contribution has been identified locally, but the transfer of scale between successive blocks has not yet been measured directly. A block-resolved intervention measuring how a scale perturbation in one block changes the next block within the same optimizer step would test the proposed depth-propagation mechanism.
- The present drift/heating decomposition is based primarily on one-step counterfactuals. These are not identical to sustained closed-loop dynamics because Adam’s second-moment adaptation changes the effective response to repeated large gradients. Matched 10–50 step branch continuations from the same checkpoints would measure this attenuation directly.
- The within-row prospective comparison currently contains only three (K, T) rows with both outcomes. The interrupted boundary-bracketing campaign, covering sixteen additional configurations, would substantially strengthen the comparison between Π , n_{eff} , and other candidate warning variables.
- A controller targeting the exact Π has not yet been tested. The present analysis predicts that such a controller should be more portable than the χ_{geo} -based controller under changes of score units and parameterization. This remains an experimental prediction.
- The metastable-escape interpretation remains qualitative. A quantitative escape parameter would require matched estimates of the restoring slope, excitation strength, and effective collapse boundary, for example quantities analogous to

$$(\lambda, D, x_c),$$

across a larger number of marginal configurations.

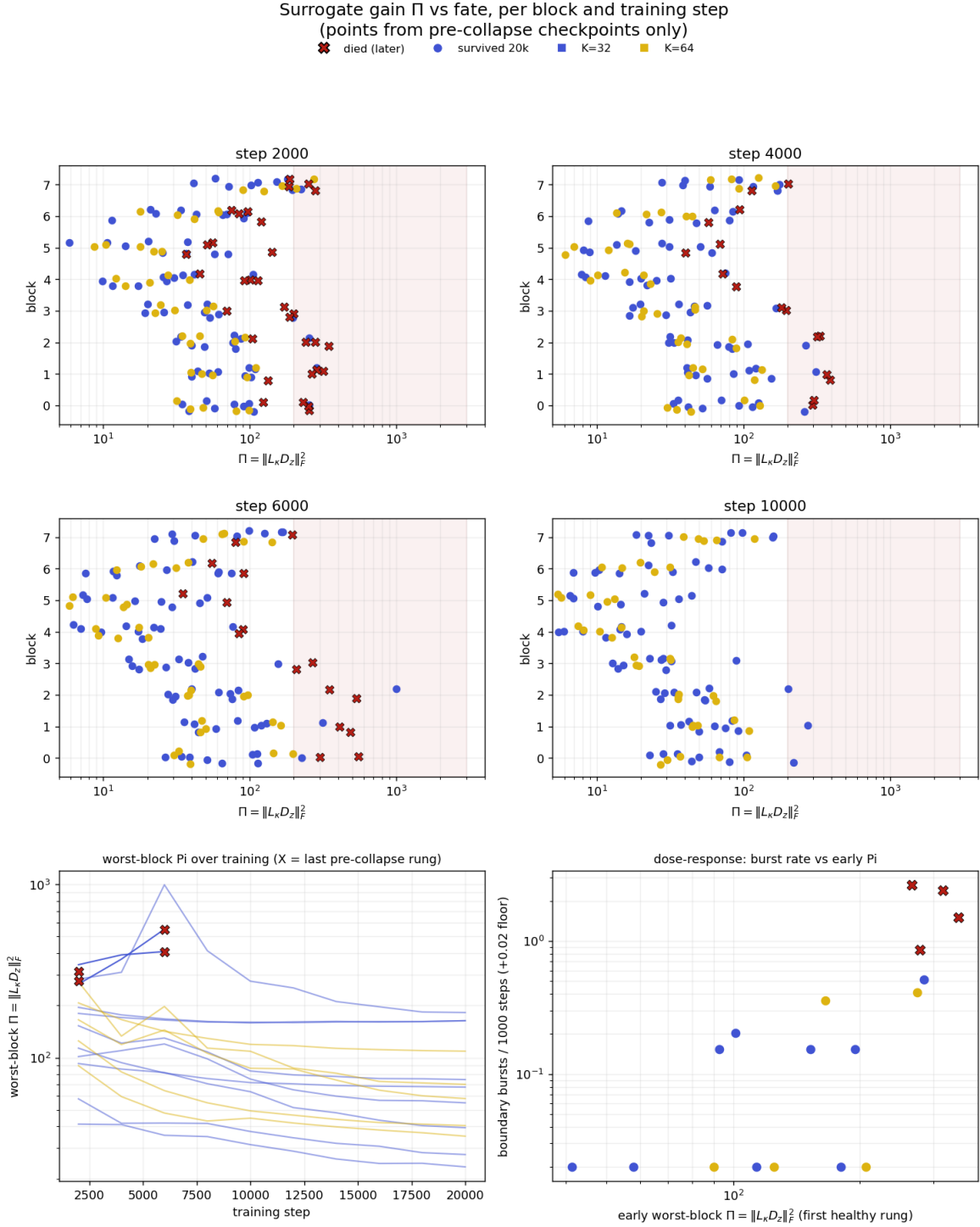


Figure 6: Exact surrogate gain Π as a function of run outcome, block, and training step. The upper panels compare surviving and eventually collapsing trajectories within individual blocks. In the lower-left panel, the worst-block Π rises toward collapse in the runs that eventually fail, while it generally decreases in surviving runs. The surviving trajectory that briefly rises and then returns corresponds to the recoverable burst in the $T = 0.5$ survivor. The lower-right panel compares pre-onset burst rate with the early worst-block value of Π .

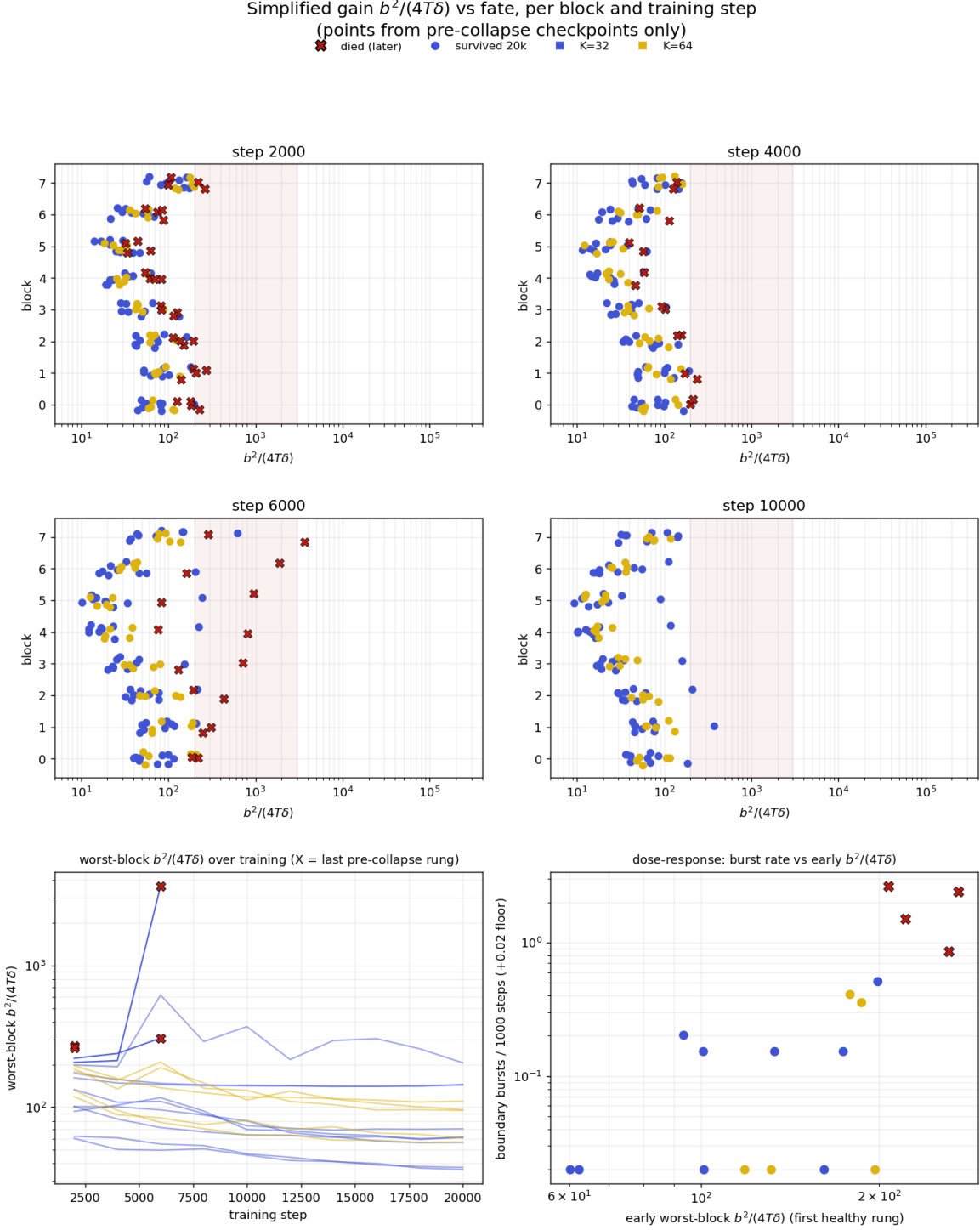


Figure 7: The same analysis using the dense-boundary approximation $b^2/(4T\delta)$. The broad dose-response ordering remains visible, but separation within individual blocks is weaker than for the exact II. Blocks with nearly vanishing local rank spacing can also produce extreme outliers because $\delta \rightarrow 0$, even when the exact surrogate gain remains finite. The exact II is therefore preferable whenever the underlying kernel tensors are available.